Q1. Kerry made some copper sulphate crystals. She wrote a description of what she did.

> **I heated some dilute sulphuric acid in a beaker and added some copper oxide to it. I stirred the mixture until it became a clear blue colour. I added more copper oxide until no more would react and then filtered the mixture into a dish. A black solid was left on the filter paper. I left the solution in the dish for a week and saw that the liquid had gone and blue crystals were left.**

Use the information in Kerry's description to answer the questions below.

(a) What colour is:

(i) copper sulphate solution?

.....….

1 mark

(ii) copper oxide?

.....….

1 mark

(b) Write down a word equation for the reaction which took place in the beaker.

..... + ... → ..... + water

1 mark

(c) Why did Kerry have to filter the mixture?

.....

.....

1 mark

Maximum 4 marks

Mark schemes

Q1.

(a) (i) blue

1

(ii) black

1

(b) copper oxide + sulphuric acid → copper sulphate + *water*

the whole word equation is required for the mark
reactants may be in either order
ignore references to oxidation states

1

(c) to separate **or** remove the copper oxide

accept 'to purify the solution'
*accept 'to separate **or** remove the black solid'*
accept 'to separate the solid from the liquid'
*do **not** accept 'because no more solid would dissolve'*

1

[4]